

ABSTRACT

Autonomous vehicles require accurate perception systems to detect surrounding objects, estimate their distance, and predict their motion. Camera-only systems provide strong appearance information but struggle with depth and velocity estimation, especially for distant objects and adverse weather conditions. This work presents CenterFusion, a radar-camera middle-fusion framework for robust 3D object detection. The approach first uses a center-based object detector to estimate preliminary 3D bounding boxes from images. A novel frustum-based radar association mechanism is then used to associate radar detections with their corresponding image objects. Radar-derived feature maps containing depth and velocity information are fused with image features to refine object properties such as depth, orientation, and velocity. Experiments on the nuScenes benchmark demonstrate significant improvements in detection quality and velocity estimation compared with camera-only approaches.

PROBLEM STATEMENT

LiDAR and camera sensor fusion has become the dominant paradigm for 3D object detection in autonomous driving systems. While LiDAR provides accurate spatial measurements and cameras offer rich appearance features, this combination has critical limitations. Both sensors are sensitive to adverse weather conditions such as fog, rain, and snow, which can severely degrade their sensing capabilities. Additionally, neither sensor can directly estimate the velocity of surrounding objects without relying on temporal information across multiple frames — a constraint that is impractical in time-critical collision avoidance scenarios.

Radar, by contrast, is highly robust to adverse weather, capable of detecting objects at ranges up to 200 meters, and uses the Doppler effect to measure object velocity instantaneously without any temporal dependency. Despite these compelling advantages, radar remains largely underutilized in modern perception pipelines. First, radar point clouds are significantly sparser than their LiDAR counterparts, making direct geometry extraction unreliable and rendering most LiDAR-based algorithms incompatible. Second, automotive radars typically report only azimuth angle and radial distance, meaning the vertical (z-axis) coordinate of each detection is either inaccurate or entirely absent, complicating 3D association with image-detected objects. Third, the data association problem — accurately linking radar detections to their corresponding objects in the scene — remains an open challenge, as multiple detections can correspond to a single object, detections may arise from background clutter, and occluded objects produce spatially ambiguous returns.

These challenges highlight the need for a sensor fusion framework that effectively combines radar and camera data for robust 3D object detection. Addressing this requires a principled solution to the radar-camera data association problem, a mechanism to compensate for radar's limited vertical accuracy, and a fusion architecture that exploits the complementary strengths of both modalities to improve depth estimation, velocity prediction, and overall detection accuracy.

BACKGROUND

3D object detection in autonomous driving has been extensively studied across single-modality and fusion-based approaches. Camera-only methods such as CenterNet extract rich appearance features but struggle with accurate depth estimation. LiDAR-based methods address this through dense 3D point clouds, using either voxel representations or 2D projections, and have achieved strong detection performance in favorable conditions. Recent fusion-based approaches have focused predominantly on combining LiDAR and camera data, exploiting their complementary strengths through early, late, or middle fusion strategies. However, radar has received comparatively little attention despite its practical advantages. The few existing radar-camera fusion methods either project radar detections directly onto the image plane — a naive approach vulnerable to z-axis inaccuracy and occlusion ambiguity — or fuse radar with LiDAR rather than camera. No dataset-rich, standardized benchmark for radar-camera fusion existed until nuScenes, which provides synchronized radar, camera, and LiDAR data for autonomous driving research.

These gaps motivate a principled middle-fusion approach that builds on the strong center-point detection framework of CenterNet, extends it with a robust radar association mechanism, and exploits radar's unique velocity and depth cues to improve 3D detection without relying on LiDAR or temporal information

KEY REQUIREMENTS

- **Resources:** Kaggle cloud computing environment with free GPU access (NVIDIA Tesla P100), no additional costs
- **Equipment & Software:** kaggle notebooks with Python, PyTorch, and standard computer vision libraries including NumPy and OpenCV
- **Access:** The nuScenes dataset is publicly available at nuscenes.org, providing synchronized radar, camera, and LiDAR data across 1000 driving scenes. The CenterNet pre-trained weights are openly accessible via the official GitHub repository.

SYSTEM MODEL

The CenterFusion system operates in two main stages: primary detection and radar-fused refinement. During primary detection, a camera image is passed through the CenterNet backbone to generate preliminary 3D bounding boxes for all detected objects, estimating their center points, depth, dimensions, and rotation using image features alone.

During the refinement stage, radar point clouds are preprocessed via pillar expansion and associated to their corresponding detected objects using the frustum association mechanism. The associated radar detections are used to generate complementary feature maps encoding depth and velocity, which are concatenated with the image features. These fused features are then passed through secondary regression heads to refine depth and rotation estimates and produce accurate velocity and attribute predictions, enabling robust 3D object detection without relying on LiDAR or temporal information.

METHODOLOGY

- **Data Preprocessing:** Raw nuScenes radar point clouds are aggregated over 3 sweeps, ego-motion compensated, and represented as 3D points $P = (x, y, z, vx, vy)$. Each radar point is then expanded into a fixed-size pillar of $[0.2, 0.2, 1.5]$ m to compensate for inaccurate height measurements. Camera images are resized from 1600×900 to 800×450 pixels, with random horizontal flipping and shifting applied during training
- **Models:**
 - **CenterNet Baseline:** Pre-trained DLA backbone detecting object center points on the image, regressing depth, dimensions, and rotation using image features only.
 - **CenterFusion with Pillar Expansion:** Radar pillars mapped directly to the image plane, filling bounding boxes with depth and velocity values as feature maps.
 - **Proposed CenterFusion (PE + Frustum Association):** Full pipeline combining pillar expansion with frustum-based radar association, generating 3-channel radar feature maps (d, vx, vy) concatenated with image features for secondary head refinement
- **Evaluation:** Performance measured using NDS, mAP, and five error metrics: mATE, mASE, mAOE, mAVE, and mAAE.

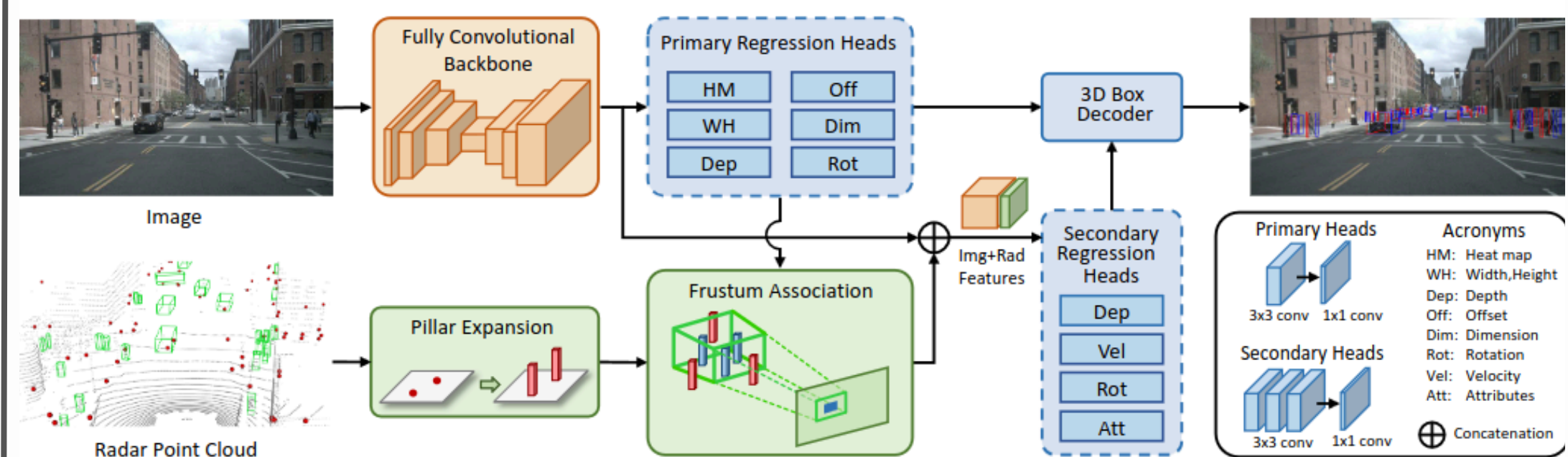


Fig1: Overview of the CenterFusion network architecture, illustrating the two-path pipeline where image features and radar point clouds are processed independently before being fused via frustum association and concatenation for refined 3D object detection.

RESULTS & DISCUSSION

- CenterFusion outperforms all camera-based baselines on the **nuScenes** benchmark, achieving an NDS of **0.449** compared to **CenterNet's 0.400** — a **12.25%** relative improvement. Velocity estimation improved dramatically, with **mAVE** reduced from **1.629** to **0.614**, a **62%** relative gain, demonstrating radar's unique contribution to motion estimation without any temporal information.

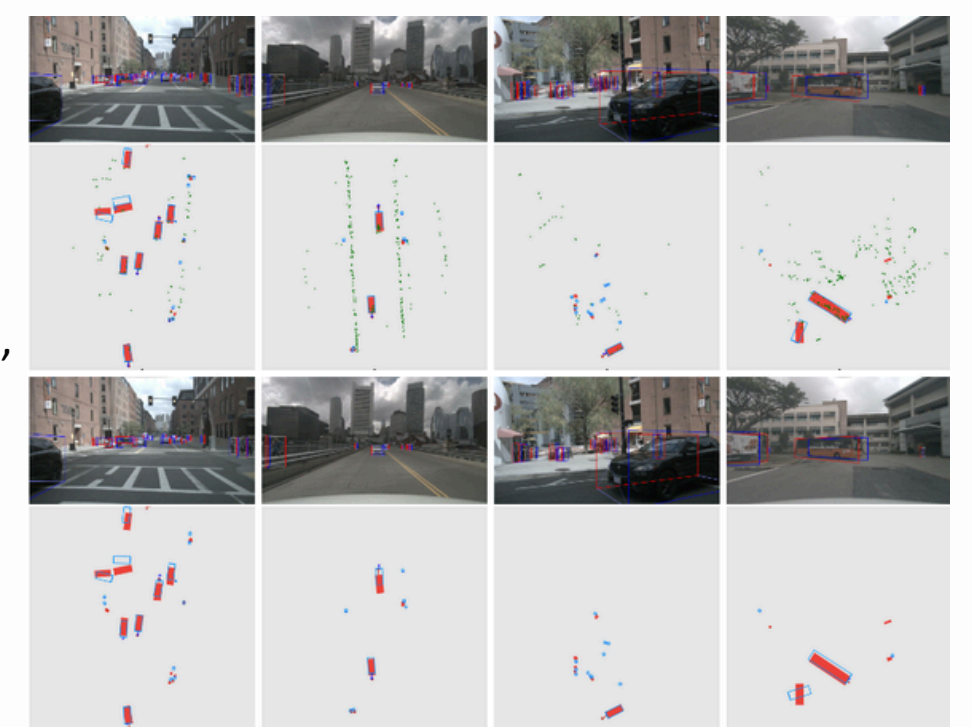


Fig2: Qualitative comparison between CenterFusion (rows 1–2) and CenterNet (rows 3–4) in both camera and Bird's Eye View (BEV). CenterFusion produces tighter 3D bounding boxes and significantly more accurate velocity predictions, particularly for distant objects.

CONCLUSION & FUTURE WORK

In summary, we proposed a new radar and camera fusion algorithm called CenterFusion, to exploit radar information for robust 3D object detection. CenterFusion accurately associates radar detections to objects on the image using a frustum-based association method, and creates radar-based feature maps to complement the image features in a middle fusion approach. Our frustum association method uses preliminary detection results to generate a RoI frustum around objects in 3D space, and maps the radar detection to the center of objects on the image. We also used a pillar expansion method to compensate for the inaccuracy in radar detections' height information, by converting radar points to fixed-size pillars in the 3D space. We evaluated our proposed method on the challenging nuScenes 3D detection benchmark, where CenterFusion outperformed the state-of-the-art camera-based object detection methods.

References

- Nabati, R., & Qi, H. (2020). CenterFusion: Center-based radar and camera fusion for 3D object detection. *arXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2011.04841>